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ENDOGENOUS GOVERNMENT SPENDING AND RICARDIAN EQUIVALENCE*

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The main implication of the Ricardian approach to budget deficits (Barro, 1974, 1989)[1] is the prediction that consumers do not spend more when they receive a deficit-financed tax cut. The argument is that rational forward-looking individuals save the additional disposable income, anticipating that the increased government debt is financed by future tax increases. Holding government spending fixed, the current tax cut is exactly offset in present value terms by higher taxes in the future. Permanent disposable income and therefore consumption remain unchanged.

Two steps are critical in this argument:

(*a*) Consumer behaviour depends only on the present value of taxes, or equivalently, on the present value of government spending plus initial debt.

(*b*) Current and future government spending is held constant, i.e. exogenous.

The present value argument, (*a*), has been debated extensively,[2] but there has been little discussion about the assumptions on government behaviour implicit in condition (*b*). But tax cuts should affect consumption if they signal changes in future government spending, as Feldstein (1982) pointed out. Predictions about government behaviour are therefore crucial, if one wants to apply the Ricardian approach as 'the benchmark model for assessing fiscal policy' (Barro, 1989).[3]

A problem in deriving such predictions is the lack of a consensus model of government behaviour. The key questions are: Is the government following arbitrary rules or is it optimising?[4] If optimising, is it maximising welfare or something else?[5] Ricardian equivalence, (*a*), combined with an arbitrary rule, namely some rule of exogenous spending, (*b*), obviously yields the striking implication that consumers should not care about budget deficits. In contrast, this paper shows that optimising government behaviour inevitably links taxes to future government spending, violating condition (*b*). Ricardian equivalence

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[1] The label 'Ricardian' is used without suggesting an interpretation of Ricardo's writings; see O'Driscoll (1977).

[2] See, e.g. Abel (1986), Barro (1989), and Bernheim (1987).

[3] In light of the debate between Feldstein (1982), Aschauer (1985, 1988), and Bernheim (1987) on how to test Ricardian equivalence in a world with uncertain government spending, I should emphasise that the paper is not a critique of Ricardian equivalence when interpreted as a statement about rational consumer behaviour. In fact, the model is set up in a way that none of the standard criticisms (e.g. involving finite lives or imperfect capital markets) applies. The question is whether its conditions will ever be satisfied under reasonable assumptions about government behaviour. See Buchanan (1976) for similar doubts about the relevance of pure financial changes and see Bohn (1991) for empirical evidence.

[4] See Lucas (1976), Sargent (1984), Sims (1986).

[5] See, e.g., Buchanan (1958), Barro (1979), Lucas and Stokey (1983).

cannot be applied. The issue here is not whether Ricardian equivalence characterises private behaviour, but whether there will ever be a Ricardian experiment of a tax cut combined with expected tax increases of equal present value.

To generate alternative predictions about government policy and its implications for consumption, the paper develops a model with welfare-maximising government and forward looking, 'Ricardian' consumers. Welfare maximisation is assumed because Ricardian theorists do not seem to object to this assumption (see Barro, 1979). But since the results are based on a simple cost-benefit tradeoff (see below), they should not be sensitive to changes in the objective function, as long as government behaviour is systematic.[6] The implications for co-movements of consumption and deficits differ dramatically from those of the Ricardian approach: The marginal propensity to consume income obtained from a tax cut can take any value between zero and the marginal propensity to consume out of ordinary income.

The basic argument why consumers react to tax cuts is as follows. A government that cares about welfare trades off marginal increases in spending on publicly provided goods against marginal reductions in private consumption. Assuming distortionary taxes (which would not cause a 'first-order' violation of Ricardian equivalence under exogenous spending), higher taxes increase the marginal cost of publicly provided goods. If consumers observe a tax-cut in this environment, they know – following the Ricardian logic – that either future taxes will have to be raised or future government spending be cut. But because of the marginal cost argument, they rationally anticipate that higher taxes will generally be accompanied by spending reductions. Thus, the Ricardian argument that lower taxes today are exactly offset by higher future taxes does not old. Consumption increases.

The implications are twofold. On the one hand, one should be cautious in simply assuming exogenous spending in deriving policy implications from Ricardian equivalence. The results are sensitive to this assumption and exogenous spending is not the result of optimising behaviour.[7] On the other hand, a positive correlation between consumption and deficits provides no evidence against forward looking consumer behaviour.

The paper is organised as follows. Section I describes the model. In Section II, the effect of fiscal policy on consumption is analysed. Section III demonstrates the link between deficits and subsequent spending policy decisions. The conclusions are reviewed in Section IV.

I. THE MODEL

Government activity and its implications for rational private behaviour will be analysed in a simple infinite horizon, representative agent economy. To shift the focus away from questions of individual behaviour, all private-sector

[6] For example, biases in spending decisions motivated by public-choice arguments could be accommodated (see footnote 10).

[7] Therefore, Lucas' (1976) critique would suggest caution even if government spending were found to be exogenous historically.

complications that would lead to non-Ricardian results in a model with exogenous government spending are omitted.[8] That is, if government spending were exogenous, the model would replicate the results of Barro (1979).

The economy consists of identical, infinitely lived individuals and a government. Each individual has preferences over two goods, c_t and g_t. In period t, individuals maximise the utility function

$$\mathrm{E}_t \sum_{i=0}^{\infty} \delta^i [u(c_{t+i}) + v(g_{t+i}, \gamma_{t+i})], \tag{1}$$

where $u(\cdot)$ and $v(\cdot)$ are concave in c and g, respectively, γ_t is a random shock to preferences, and $0 < \delta < 1$ is the rate of time preference. Individuals buy the good c_t privately, but good g_t is provided by the government in equal quantity to all individuals.[9] The shock γ_t is intended to reflect changing preferences for the publicly provided good; therefore, assume $v_{g\gamma} \neq 0$.

Individuals take government decisions about taxation and spending as given. They have initial wealth (debt and capital) and earn after-tax income y_t (defined below) to pay for the private good and to invest in capital, K_t, and government bonds, D_t.

Each unit of capital K_t yields $R \equiv 1+r > 1$ units of goods in period $t+1$. To simplify, assume that the interest rate r is constant, that the rate of time discount equals the interest rate, i.e. $\delta R = 1$, and that the government can commit itself not to tax capital (to eliminate time consistency issues). Then government bonds and capital are perfect substitutes. The individual budget constraint in period t is

$$c_t + K_t + D_t = R(K_{t-1} + D_{t-1}) + y_t. \tag{2}$$

The government uses taxation and debt financing to pay for the publicly provided good g_t. It chooses spending and taxes to maximise welfare, i.e., the same utility function (1).[10] Without loss of generality, units are defined so that c_t and g_t have identical production cost per capita and that prices are normalised to one. Given initial government debt, the budget constraint in period t is

$$g_t + RD_{t-1} = T_t + D_t, \tag{3}$$

where T_t denotes tax revenues.

Since lump-sum taxes are not widely used in practice, taxes are allowed to have distortionary effects. (This feature in itself does not cause real effects of tax

[8] Examples of such complications are capital market imperfections, finite lives and different bequest motives or uncertainty about the incidence of future taxes. See, e.g., Abel (1986), Barro (1979, 1981, 1989), Barsky *et al.* (1986), Carmichael (1982), Chan (1983), and the recent surveys by Aschauer (1988) and Bernheim (1987). To obtain unique policies, distortionary taxes are allowed, but they are introduced in a way that would yield Ricardian results in the presence of exogenous spending, as in Barro (1979); see Corollary 2 below.

[9] This good may be a public good in the sense that no person can be excluded from consuming it. Then taxation arises naturally as solution of the free rider problem. But it may be any other good or service that is financed by taxes. Separability is only imposed to eliminate inessential interaction between consumption and government spending choices. Bohn (1990) shows that the results of the theorems below generalise to the case of non-separable preferences.

[10] Readers who have different beliefs about the government objective function (cf. Buchanan, 1958) may interpret the $v(\cdot)$-part of (1) as whatever determines government's choice of public spending.

policy; see Corollary 2 below.) Denote the exogenous, stochastic income that would be realised if there were no taxes by Y_t. If the government raises revenue T_t, it imposes a burden on individuals that exceeds T_t by $h(T_t, Y_t, \epsilon_t)$, so that disposable income is

$$y_t = y(T_t, Y_t, \epsilon_t) = Y_t - T_t - h(T_t, Y_t, \epsilon_t). \tag{4}$$

It is assumed that the excess burden $h(\cdot)$ is an increasing convex function of tax revenue, and that there is a stochastic shock ϵ_t, which increases the marginal cost of raising revenue, i.e. $h_T > 0$, $h_{TT} > 0$, and $h_{Te} \neq 0$. These assumptions on excess burden closely follow Barrow (1979). They may be viewed as a reduced form solution of the individuals' problem in a model with taxes on labour income.[11]

An advantage of summarising excess burden in this form is that the function $h(\cdot)$ captures all aspects of self-interested private behaviour that may reduce social welfare, such as efforts to avoid individual tax payments. Therefore, the equilibrium allocation can be obtained by solving a social planner's problem in which the government is allowed to choose consumption in addition to its policy variables, but takes the loss function $h(T_t, Y_t, \epsilon_t)$ as given.

The solution of the social planning problem describes the paths of all macroeconomic variables as functions of the stochastic shocks. Movements in policy variables in response to shocks can be interpreted as unexpected policy changes. In this way, one can do essentially comparative static analysis in a model with optimal policy. If there were no shocks, all variables would evolve deterministically as predicted by the solution to the social planning problem at some initial date. Policy experiments – comparative statics – would be logically inconsistent with the model. With uncertainty, however, unexpected policy changes occur whenever the social planner re-optimises in response to some stochastic shocks. The effects of such policy changes on the economy can be studied. In contrast to standard comparative static analysis, policy will always be optimal for the given set of shocks. Private behaviour will be optimal for the given shocks and the given policy.

Unfortunately, a lengthy and complicated stochastic dynamic programming approach is required to characterise the outcomes of the social planning problem in general. But since the stochastic shocks γ_t, Y_t, and ϵ_t are only needed to generate policy changes (in a way consistent with optimisation), it is sufficient to concentrate on stochastic shocks in one particular period, say $t = 0$, and to abstract from later stochastic disturbances. That is, assume

$$\gamma_t = 0, Y_t = \bar{Y} \quad \text{and} \quad \epsilon_t = 0 \quad \text{for all} \quad t \geqslant 1, \tag{5}$$

where $\bar{Y}$ is a constant. The general results without these simplifying assumptions are qualitatively identical and they are proven in Bohn (1990). Here, I will concentrate on the simpler case where (5) is assumed. To simplify notation, also assume that initial values of debt and capital are $D_{-1} = K_{-1} = 0$. Shocks in

[11] See Bohn (1990). A key restriction implied by this specification is that taxation causes distortions within a period, but not intertemporally. It is well known that tax induced distortions may take a much more complicated form and they may provide alternative justifications for real effects of deficits (see Judd 1985*a*, *b*; 1987*a*, *b*). But such complications would be distracting here.

period $t = 0$ generate alternative policies that will be compared. From then on, the economy evolves deterministically and can be solved for any combination of period-0 shocks.

The social planner's problem in period zero is then to maximise utility (1) subject to constraints (2)–(4). The resulting first order conditions are

$$u_c(c_0) = u_c(c_t) \tag{6a}$$

$$v_g(g_0, \gamma_0) = v_g(g_t) \tag{6b}$$

$$v_g(g_0, \gamma_0) - [1 + h_T(T_0, Y_0, \epsilon_0)]\, u_c(c_0) = 0 \tag{6c}$$

$$v_g(g_t) - [1 + h_T(T_t)]\, u_c(c_t) = 0 \tag{6d}$$

for all periods $t \geqslant 1$, where constant arguments (due to (5)) are omitted. Since equation (6a) is the first order condition for individuals maximising (1) subject to (2), the social planning problem solves the game between government and individuals. Equations (6a, b) determine the intertemporal allocation of consumption and government spending. Equations (6c, d) characterised the tradeoff between private consumption and publicly provided goods. Since equation (6a) implies constant consumption for all periods and equations (6b, d) imply constant government spending and taxes for all periods $t \geqslant 1$, define $c \equiv c_t = c_0$, $g \equiv g_t$, and $T \equiv T_t$ for all $t \geqslant 1$. Then the budget constraints (2) and (3) simplify to

$$(1+r)\, c = r[Y_0 - T_0 - h(T_0, Y_0, \epsilon_0)] + [\bar{Y} - T - h(T)], \tag{7a}$$

$$rg_0 + g = rT_0 + T, \tag{7b}$$

respectively, which provides an explicit solution for c. Links between government policy and consumption in this setting are analysed in the next section.

II. FISCAL POLICY AND CONSUMPTION

Equation (6) and (7) determine a unique solution for (c, g_0, T_0, g, T) for each possible set of realisations of $(Y_0, \epsilon_0, \gamma_0)$. An econometrician may draw different realisations of shocks $(Y_0, \epsilon_0, \gamma_0)$ to study the joint movements of consumption and government policy.[12] Since the expected values of $(Y_0, \epsilon_0, \gamma_0)$ are $(\bar{Y}, 0, 0)$, deviations of government spending and taxes from their values under $(\bar{Y}, 0, 0)$ can be interpreted as unexpected changes in policy.

In order to allow independent variation in the three key variables c, g_0, and T_0, at least three sources of variability are necessary. Here, each of the three shocks $(Y_0, \epsilon_0, \gamma_0)$ affects all endogenous variables, but not equally strongly. Since the main effect of ϵ_0 is on T_0 observed variations in T_0 can be interpreted as largely caused by changes in ϵ_0. Similarly, changes Y_0 move c and changes in γ_0 strongly affect g_0. The main issue is whether – and if so, how much – consumers react to debt-financed tax cuts, i.e., how consumption varies with a

[12] Because shocks occur only once, an observer would need a cross section of economies to draw inferences. In the general case with repeated shocks, a time series on one economy would be sufficient, provided changing initial conditions are taken into account. Bohn (1990) shows that the theoretical results stated below under the simplifying assumption (5) generalise to the case of repeated shocks.

set of shocks that reduce taxes T_0 and just leave government spending g_0 unchanged. Such a causal interpretation of co-movements between consumption and policy variables is possible, because changes in consumption can be decomposed into changes directly due to shocks and changes due to tax policy, as follows.

Equation (7*a*) shows that consumption depends only on the present value of disposable income (recall the definition (4)), where period-0 disposable income is a function of shocks and taxes and future disposable income depends only on future taxes, T. If one defines 'before-tax income' $y^* = Y_0 - h(\bar{T}, Y_0, \epsilon_0)$, for some initial value $\bar{T}$, marginal changes in disposable income, y_0, can be decomposed into changes caused by taxes and changes due to shocks that enter through y^*:

$$dy_0 = -(1+h_T)\,dT_0 + dy^*. \tag{8}$$

Thus, if consumption varies with taxes T_0 after controlling for changes in y^*, one may conclude that policy affects consumption. If the effect remains even with unchanged current governments spending, changes in current taxes are not offset in present value terms by future changes in taxes, i.e., they must signal future changes in government spending.

The magnitude of policy-induced consumption changes can be evaluated in comparison to a change in disposable income that is not accompanied by policy changes. Therefore, define the (ordinary) marginal propensity to consume (MPC) as

$$MPC_Y = \left.\frac{dc_0}{dy_0}\right|_{dg_0=0,\, dT_0=0}$$

and define the marginal propensity to consume income obtained from a tax cut as

$$MPC_T = \left.\frac{dc_0}{dy_0}\right|_{dg_0=0,\, dy^*=0}.$$

Both MPCs have a regression interpretation. MPC_T and MPC_Y would be the coefficients on taxes and before-tax income, y^*, respectively, in a regression of consumption on these two variables, a constant, and government spending. Very similar regressions have been run by Feldstein (1982) and others (see the survey by Bernheim, 1987). The following result relates the two MPCs to the fundamental parameters of the models:

THEOREM 1. *Define* $\Delta = (1+r)\,[h_{TT}\,u_c + (-v_{gg})] + (1+h_T)^2(-u_{cc}) > 0$. *Then,*

$$MPC_Y = (r/\Delta)\,[h_{TT}\,u_c + (-v_{gg})],$$
$$MPC_T = (r/\Delta)\,h_{TT}\,u_c.$$

Proof. Marginal changes in c_0, T_0, y^* are obtained by taking the total differential of (6), (7), and (8). The MPCs are ratios of pairs of these changes. See Bohn (1990) for a more detailed proof.

COROLLARY 1 (*General Case*). *If* h_{TT}, $(-v_{gg})$, *and* $(-u_{cc})$ *are strictly positive and finite, then* $0 < MPC_T < MPC_Y < r/(1+r)$.

COROLLARY 2 (*A 'Ricardian' Result*). *As* $-v_{gg}(g) \to \infty$ *at some value of* g, *consumption does not vary with taxes:* $MPC_T \to 0$.

COROLLARY 3. *As* $-v_{gg}(g) \to 0$, *consumers do not distinguish receipts from tax-cuts from other income:* $MPC_T \to MPC_Y$.

One should note that all the qualitative results – but not the exact formulas in Theorem 1 – generalise to environments without assumption (5) and to environments without separability between goods c and g (see Bohn, 1990). Thus, the reaction of consumption to tax cuts is generally positive but less than the reaction to other changes in disposable income. The key equation for understanding the results is (6*d*), which requires that the marginal rate of substitution between privately and publicly purchased goods equals their relative price:

$$\frac{v_g(g)}{u_c(c)} = 1 + h_T(T). \tag{9}$$

The relative price of publicly provided goods is unity (production cost) plus the marginal cost of distortions h_T. Higher taxes increase the marginal cost of g and induce substitution from good g to the private good c. Because of this negative link between government spending and taxes, the budgetary adjustment following a period-0 tax cut generally involves both higher taxes and lower government spending. Consumers rationally anticipate this adjustment. Because of lower anticipated government spending, a tax cut unambiguously increases consumption. But because of higher future taxes, they react less to a tax cut than to higher income from other sources.

If $(-v_{gg}) \to \infty$ at some value of g (Corollary 2), government spending is fixed at that value. Then a tax cut does not change the present value of taxes and therefore leaves consumption unchanged. Since the government behaves as the Ricardian approach assumes, Ricardian equivalence emerges. Interestingly, this holds even though taxes are distortionary.[13]

If $(-v_{gg}) = 0$ (Corollary 3), demand for publicly provided goods is infinitely elastic to changes in relative price. Different levels of period-0 debt do not lead to differences in taxes later on. Therefore, individuals correctly view any period-zero tax cut as increase in net wealth and spend it like any other income.[14]

In general, consumer reactions are in between those of Corollaries 2 and 3.[15] If demand for the good g is very sensitive to cost (v_{gg} small in absolute value), higher debt will not lead to future tax increases but rather to future reductions

[13] The reason is that optimal policy equalises excess burden over time, $h_T(T_0, Y_0, E_0) = h_T(T_t)$ for all t, as in Barro (1979). Barro's (1989) claim that distortionary taxes induce only second-order effects is confirmed in this model. Of course, distortionary taxes may make a difference in other models; see Judd (1985*a*, *b*; 1987*a*, *b*) for detailed analyses.

[14] Notice, though, that the MPC is less than $r/(1+r)$, because any increase in net wealth is allocated to both goods, leading to somewhat higher taxes later. In contrast, the Ricardian case has $MPC_Y = r/(1+r)$, because spending on public goods does not increase even if private wealth rises.

[15] Since the focus of this paper is on exogeneity of government spending, only two limiting cases are explored. Other limiting results can be derived easily, e.g., Ricardian equivalence for $h_{TT}(\cdot) \to 0$, which replicates Barro (1974).

in spending. Then the result is far from the Ricardian prediction. On the other hand, Ricardian equivalence will be a good benchmark, if government spending is very inelastic ($(-v_{gg})$ large).

Another feature of the optimising model is interesting: When aggregate income increases, consumption rises by less than $r/(1+r)$, the MPC in the standard permanent income model. The reason for the lower MPC_Y is that increases aggregate income is optimally allocated to both goods. Therefore, whenever income increases on aggregate, a fraction $\{1-[(1+r)/r]\,MPC_Y\} > 0$ must be set aside for future taxes.[16]

III. A NET WEALTH INTERPRETATION

To understand consumer behaviour in the welfare-maximising model, it is useful to reexamine Barro's (1974) famous question of what constitutes net wealth. Period-0 events influence the future through debt, D_0, and capital, K_0, which determine initial conditions at $t = 1$. One can show:

THEOREM 2. *Taxes in periods* $t \geqslant 1$ *are a function of initial debt and capital,* $T = T(D_0, K_0)$, *with derivatives*

$$T_K = r(1+h_T)(-u_{cc})/A,$$
$$T_D = r[-v_{gg}-(1+h_T)\,u_{cc}]/A,$$

where $A = -v_{gg}-(1+h_T)^2 u_{cc}+h_{TT}\,u_c > 0.$

Proof. By taking the total differential of ($6d$) or (9) subject to the budget constraints.

For given capital, higher government debt that is held by individuals leads to higher taxes. But since $T_D < r$, taxes do not increase enough to pay off the debt at constant government spending. In this model, the answer to Barro's question is that the fraction $(1-T_D/r)$ of government bonds are indeed net wealth. In addition, higher aggregate capital, K_0, leads to somewhat higher taxes (since $0 < T_K < T_D < r$), which are used to increase the availability of good g. Thus, not all of aggregate capital is net wealth either, which motivates the result $MPC_Y < r/(1+r)$ above.

IV. CONCLUSIONS

The paper shows that an optimising model of government behaviour has implications that can differ radically from those of the Ricardian approach, even though both approaches assume rational, forward-looking individuals. Both agree that a tax cut has a smaller effect on consumption than other increases in disposable income because some increase in future taxes is expected. But the optimising model predicts also lower future government spending which reduces the need for higher taxes. The magnitude of the

[16] This argument does not involve taxes on capital that would distort individual saving decisions. If one introduced idiosyncratic shocks to income, individuals would indeed consume $r/(1+r)$ of such increases.

spending change depends on how sensitive the demand for public goods is to cost. With high elasticity, individuals will spend almost as much income from tax cuts as they spend from other disposable income. With zero elasticity, Ricardian equivalence emerges.

The policy implications are wide-ranging. Unless the assumption of exogenous government spending is true, correlation between consumption and deficits provides no evidence against forward looking consumer behaviour. Research on the determinants of government spending is needed to determine how rational consumers should react to tax cuts. Perhaps the way in which the current 'deficit-problems' in the United States, Germany, and several other countries are resolved – by spending cuts and/or by tax increases – will provide some revealing experiments.

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